Input system:
Dependent variables:$r_1, x_2, p_2, q_3, p_3, q_1, x_1, y_1, z_2, r_2, r_3, y_2, p_1, z_3, y_3, x_3, z_1, q_2$
Independent variables:t
Equations:

$$\frac{dx_1}{dt} = p_1$$

$$\frac{dy_1}{dt} = r_1$$

$$\frac{dz_1}{dt} = q_1$$

$$\frac{dx_2}{dt} = p_2$$

$$\frac{dy_2}{dt} = r_2$$

$$\frac{dz_2}{dt} = q_2$$

$$\frac{dx_3}{dt} = p_3$$

$$\frac{dy_3}{dt} = r_3$$

$$\frac{dz_3}{dt} = q_3$$

$$\begin{aligned}\frac{dp_1}{dt} &= Gm_2(x_2 - x_1)\text{isqrt}((x_1 - x_2)^2 + (y_1 - y_2)^2 + (z_1 - z_2)^2)^3 \\ &\quad + Gm_3(x_3 - x_1)\text{isqrt}((x_1 - x_3)^2 + (y_1 - y_3)^2 + (z_1 - z_3)^2)^3\end{aligned}$$

$$\begin{aligned}\frac{dr_1}{dt} &= Gm_2(y_2 - y_1)\text{isqrt}((x_1 - x_2)^2 + (y_1 - y_2)^2 + (z_1 - z_2)^2)^3 \\ &\quad + Gm_3(y_3 - y_1)\text{isqrt}((x_1 - x_3)^2 + (y_1 - y_3)^2 + (z_1 - z_3)^2)^3\end{aligned}$$

$$\begin{aligned}\frac{dq_1}{dt} &= Gm_2(z_2 - z_1)\text{isqrt}((x_1 - x_2)^2 + (y_1 - y_2)^2 + (z_1 - z_2)^2)^3 \\ &\quad + Gm_3(z_3 - z_1)\text{isqrt}((x_1 - x_3)^2 + (y_1 - y_3)^2 + (z_1 - z_3)^2)^3\end{aligned}$$

$$\begin{aligned}\frac{dp_2}{dt} &= Gm_1(x_1 - x_2)\text{isqrt}((x_1 - x_2)^2 + (y_1 - y_2)^2 + (z_1 - z_2)^2)^3 \\ &\quad + Gm_3(x_3 - x_2)\text{isqrt}((x_2 - x_3)^2 + (y_2 - y_3)^2 + (z_2 - z_3)^2)^3\end{aligned}$$

$$\frac{dr_2}{dt} = Gm_1(y_1 - y_2)\text{isqrt}((x_1 - x_2)^2 + (y_1 - y_2)^2 + (z_1 - z_2)^2)^3 + Gm_3(y_3 - y_2)\text{isqrt}((x_2 - x_3)^2 + (y_2 - y_3)^2 + (z_2 - z_3)^2)^3$$

$$\frac{dq_2}{dt} = Gm_1(z_1 - z_2)\text{isqrt}((x_1 - x_2)^2 + (y_1 - y_2)^2 + (z_1 - z_2)^2)^3 + Gm_3(z_3 - z_2)\text{isqrt}((x_2 - x_3)^2 + (y_2 - y_3)^2 + (z_2 - z_3)^2)^3$$

$$\frac{dp_3}{dt} = Gm_1(x_1 - x_3)\text{isqrt}((x_1 - x_3)^2 + (y_1 - y_3)^2 + (z_1 - z_3)^2)^3 + Gm_2(x_2 - x_3)\text{isqrt}((x_2 - x_3)^2 + (y_2 - y_3)^2 + (z_2 - z_3)^2)^3$$

$$\frac{dr_3}{dt} = Gm_1(y_1 - y_3)\text{isqrt}((x_1 - x_3)^2 + (y_1 - y_3)^2 + (z_1 - z_3)^2)^3 + Gm_2(y_2 - y_3)\text{isqrt}((x_2 - x_3)^2 + (y_2 - y_3)^2 + (z_2 - z_3)^2)^3$$

$$\frac{dq_3}{dt} = Gm_1(z_1 - z_3)\text{isqrt}((x_1 - x_3)^2 + (y_1 - y_3)^2 + (z_1 - z_3)^2)^3 + Gm_2(z_2 - z_3)\text{isqrt}((x_2 - x_3)^2 + (y_2 - y_3)^2 + (z_2 - z_3)^2)^3$$

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Result:

Dependent variables:$r_1, x_2, p_3, q_1, s_1, y_1, z_2, r_2, r_3, y_2, p_1, y_3, x_3, x_1, p_2, q_3, s_3, z_3, s_2, q_2, z_1$

Independent variables:t

Additional variables:

$$s_1 = \text{isqrt}((x_1 - x_2)^2 + (y_1 - y_2)^2 + (z_1 - z_2)^2)$$

$$s_2 = \text{isqrt}((x_1 - x_3)^2 + (y_1 - y_3)^2 + (z_1 - z_3)^2)$$

$$s_3 = \text{isqrt}((x_2 - x_3)^2 + (y_2 - y_3)^2 + (z_2 - z_3)^2)$$

Equations:

$$\frac{dx_1}{dt} = p_1$$

$$\frac{dy_1}{dt} = r_1$$

$$\frac{dz_1}{dt} = q_1$$

$$\frac{dx_2}{dt} = p_2$$

$$\frac{dy_2}{dt} = r_2$$

$$\frac{dz_2}{dt} = q_2$$

$$\frac{dx_3}{dt} = p_3$$

$$\frac{dy_3}{dt} = r_3$$

$$\frac{dz_3}{dt} = q_3$$

$$\frac{dp_1}{dt} = s_1^3 G m_2 x_2 + (-1) s_1^3 G m_2 x_1 + s_2^3 G m_3 x_3 + (-1) s_2^3 G m_3 x_1$$

$$\frac{dr_1}{dt} = s_1^3 G m_2 y_2 + (-1) s_1^3 G m_2 y_1 + s_2^3 G m_3 y_3 + (-1) s_2^3 G m_3 y_1$$

$$\frac{dq_1}{dt} = s_1^3 G m_2 z_2 + (-1) s_1^3 G m_2 z_1 + s_2^3 G m_3 z_3 + (-1) s_2^3 G m_3 z_1$$

$$\frac{dp_2}{dt} = s_1^3 G m_1 x_1 + (-1) s_1^3 G m_1 x_2 + s_3^3 G m_3 x_3 + (-1) s_3^3 G m_3 x_2$$

$$\frac{dr_2}{dt} = s_1^3 G m_1 y_1 + (-1) s_1^3 G m_1 y_2 + s_3^3 G m_3 y_3 + (-1) s_3^3 G m_3 y_2$$

$$\frac{dq_2}{dt} = s_1^3 G m_1 z_1 + (-1) s_1^3 G m_1 z_2 + s_3^3 G m_3 z_3 + (-1) s_3^3 G m_3 z_2$$

$$\frac{dp_3}{dt} = s_2^3 G m_1 x_1 + (-1) s_2^3 G m_1 x_3 + s_3^3 G m_2 x_2 + (-1) s_3^3 G m_2 x_3$$

$$\frac{dr_3}{dt} = s_2^3 G m_1 y_1 + (-1) s_2^3 G m_1 y_3 + s_3^3 G m_2 y_2 + (-1) s_3^3 G m_2 y_3$$

$$\frac{dq_3}{dt} = s_2^3 G m_1 z_1 + (-1) s_2^3 G m_1 z_3 + s_3^3 G m_2 z_2 + (-1) s_3^3 G m_2 z_3$$

$$\begin{aligned}\frac{ds_1}{dt} = {} & (-1) s_1^3 p_1 x_1 + s_1^3 p_1 x_2 + s_1^3 p_2 x_1 + (-1) s_1^3 p_2 x_2 + (-1) s_1^3 r_1 y_1 + s_1^3 r_1 y_2 \\ & + s_1^3 r_2 y_1 + (-1) s_1^3 r_2 y_2 + (-1) s_1^3 q_1 z_1 + s_1^3 q_1 z_2 + s_1^3 q_2 z_1 + (-1) s_1^3 q_2 z_2\end{aligned}$$

$$\begin{aligned}\frac{ds_2}{dt} = {} & (-1) s_2^3 p_1 x_1 + s_2^3 p_1 x_3 + s_2^3 p_3 x_1 + (-1) s_2^3 p_3 x_3 + (-1) s_2^3 r_1 y_1 + s_2^3 r_1 y_3 \\ & + s_2^3 r_3 y_1 + (-1) s_2^3 r_3 y_3 + (-1) s_2^3 q_1 z_1 + s_2^3 q_1 z_3 + s_2^3 q_3 z_1 + (-1) s_2^3 q_3 z_3\end{aligned}$$

$$\begin{aligned}\frac{ds_3}{dt} = {} & (-1) s_3^3 p_2 x_2 + s_3^3 p_2 x_3 + s_3^3 p_3 x_2 + (-1) s_3^3 p_3 x_3 + (-1) s_3^3 r_2 y_2 + s_3^3 r_2 y_3 \\ & + s_3^3 r_3 y_2 + (-1) s_3^3 r_3 y_3 + (-1) s_3^3 q_2 z_2 + s_3^3 q_2 z_3 + s_3^3 q_3 z_2 + (-1) s_3^3 q_3 z_3\end{aligned}$$